ASCEND-PG

Evaluation Parameters Reference

Complete list of all metrics, KPIs, and parameters across the system

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1. Attack Metrics (RL Agent)

1.1 Evasion Metrics

Parameter	Type	Range	Description	Source
evasion_rate	float	[0, 1]	Percentage of steps that evaded detection	rl_env.py
recent_evasion_rate	float	[0, 1]	Sliding window (last 5 steps) evasion	rl_env.py
detected	bool	T/F	Whether current step was detected	rl_env.py
detection_confidence	float	[0.0, 1.0]	Detection confidence score per step	rl_env.py
detection_method	string	-	Detection mechanism used	rl_env.py

1.2 Campaign Metrics

Parameter	Type	Range	Description	Source
campaign_complete	bool	T/F	All 4 tactical stages completed	rl_env.py
completed_stages	set	{0,1,2,3}	Which stages were completed	rl_env.py
chain	list	-	List of technique IDs used	rl_env.py
step_count	int	0-16	Steps in current episode	rl_env.py

1.3 Targeting Metrics

Parameter	Type	Range	Description	Source
target_bus	int	[0, N)	Which bus was attacked	rl_env.py
compromised_buses	set	-	Set of compromised bus indices	rl_env.py
attack_depth	float	[0, 1]	Min hop distance to generators	rl_env.py
frac_compromised	float	[0, 1]	Fraction of buses compromised	rl_env.py

1.4 Action Parameters

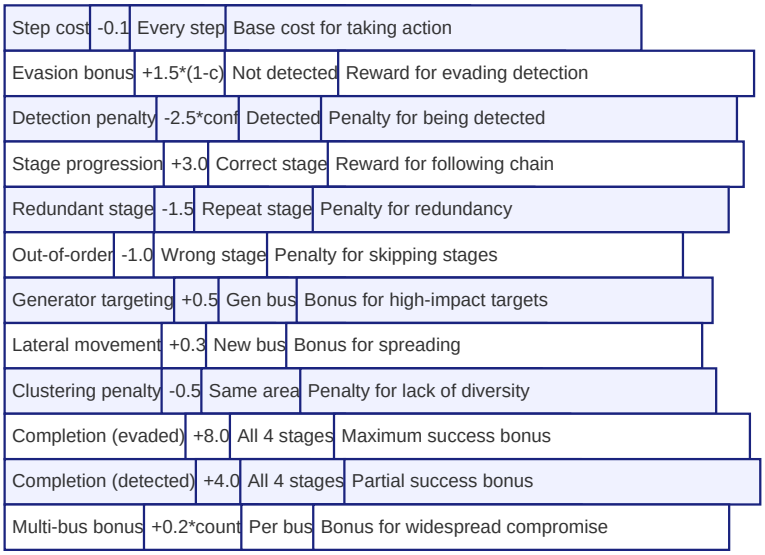
Parameter	Type	Range	Description	Source
register_address	int	40000-40049	Modbus register address	rl_env.py
register_offset	int	[0, 49]	Offset from base register	rl_env.py
value_magnitude	float	[0.9, 1.1]	Write value scale factor	rl_env.py
value_level	int	[0, 9]	Value magnitude index	rl_env.py

1.5 Detection Risk Multipliers

Parameter	Formula	Range	Description	Source
register_risk	1-(off/50)*0.3	0.7-1.0	Risk from register choice	rl_env.py
value_deviation	abs(val-1.0)	0.0-0.1	Deviation from normal	rl_env.py
value_risk	1+dev*5.0	1.0-1.5	Risk from value deviation	rl_env.py
target_risk	1.2 if gen	1.0-1.2	Risk from target type	rl_env.py
risk_multiplier	reg*val*tgt	0.7-1.8	Combined detection risk	rl_env.py

1.6 Reward Function Components

Component	Value	Trigger	Description


Step cost -0.1 Every step Base cost for taking action
Evasion bonus +1.5*(1-c) Not detected Reward for evading detection
Detection penalty -2.5*conf Detected Penalty for being detected
Stage progression +3.0 Correct stage Reward for following chain
Redundant stage -1.5 Repeat stage Penalty for redundancy
Out-of-order -1.0 Wrong stage Penalty for skipping stages
Generator targeting +0.5 Gen bus Bonus for high-impact targets
Lateral movement +0.3 New bus Bonus for spreading
Clustering penalty -0.5 Same area Penalty for lack of diversity
Completion (evaded) +8.0 All 4 stages Maximum success bonus
Completion (detected) +4.0 All 4 stages Partial success bonus
Multi-bus bonus +0.2*count Per bus Bonus for widespread compromise

2. Defense Metrics

2.1 Hybrid Detector

Parameter	Type	Range	Description	Source
suricata_detected	bool	T/F	Suricata found matching alerts	hybrid_detector
suricata_alert_count	int	0+	Number of matching alerts	hybrid_detector
suricata_signatures	list	-	List of matched alert signatures	hybrid_detector
ml_detected	bool	T/F	ML fusion exceeded threshold	hybrid_detector
ml_confidence	float	[0, 1]	Fused ML confidence score	hybrid_detector
ml_lstm_confidence	float	[0, 1]	LSTM detector confidence	hybrid_detector
ml_ae_score	float	[0, 1]	Autoencoder anomaly score	hybrid_detector
ml_available	bool	T/F	ML models loaded and working	hybrid_detector
combined_detected	bool	T/F	Overall OR of suricata + ml	hybrid_detector
detection_method	string	-	both/suricata/ml/undetected	hybrid_detector

2.2 ML Detector

Parameter	Type	Range	Description	Source
lstm_confidence	float	[0, 1]	LSTM sigmoid output	ml_detector
ae_anomaly_score	float	[0, 1]	Normalized AE reconstruction error	ml_detector
ae_max_recon_error	float	0+	Raw max MSE across window	ml_detector
fused_confidence	float	[0, 1]	Weighted avg of LSTM + AE	ml_detector
detected	bool	T/F	fused_confidence > 0.50	ml_detector

2.3 ML Training Metrics

Metric	Formula	Description
accuracy	(TP+TN)/Total	Overall classification accuracy
precision	TP/(TP+FP)	Of flagged-as-attack, how many truly are
recall	TP/(TP+FN)	Of real attacks, how many were caught
f1	2*P*R/(P+R)	Harmonic mean of precision and recall
tp	-	True Positives
fp	-	False Positives
fn	-	False Negatives
tn	-	True Negatives

2.4 Suricata IDS Rules

SID	Rule Name	Technique	Description
9000001	FDI - Holding Register Write	T0836	False data injection
9000002	DoS - Modbus Request Flood	T0814	Denial of service
9000004	Recon - Device ID Query	T0888	Device enumeration
9000005	Recon - Read Exception Status	T0840	Reconnaissance
9000006	Recon - Diagnostics FC08	T0840	Diagnostics scan

9000007	CmdInject - Coil Write	T0855	Command injection
9000008	CmdInject - Diagnostics Restart	T0855	Restart attempt
9000009	CmdInject - Clear Counters	T0855	Counter tampering
9000010	DoS - Rapid TCP SYN	T0814	SYN flood
9000020	DoS - Sub-Threshold Read Flood	T0814	Slow DoS
9000022	MITM - Non-Standard Port 503	T0830	MITM attempt
9000023	BruteForce - Rapid Register Enum	T0806	Brute force
9000024	DoV - Forged FC06 Write	T0815	Denial of view
9000025	Recon - Coil Read	T0801	Coil reconnaissance

3. Network Metrics

3.1 Flow Logger Metrics

Parameter	Type	Unit	Description	Source
duration_sec	float	seconds	Flow duration	flow_logger
total_packets	int	packets	Total packets in flow	flow_logger
total_bytes	int	bytes	Total bytes in flow	flow_logger
fwd_packets	int	packets	Forward direction count	flow_logger
bwd_packets	int	packets	Backward direction count	flow_logger
fwd_bytes	int	bytes	Forward direction bytes	flow_logger
bwd_bytes	int	bytes	Backward direction bytes	flow_logger
bytes_per_sec	float	B/s	Throughput	flow_logger
packets_per_sec	float	pkt/s	Packet rate	flow_logger
syn_count	int	count	SYN flag count	flow_logger
ack_count	int	count	ACK flag count	flow_logger
rst_count	int	count	RST flag count	flow_logger
fin_count	int	count	FIN flag count	flow_logger

3.2 CIC-IDS-2017 Features (78 dimensions)

Category	Count	Features
Port & Duration	2	Destination Port, Flow Duration
Packet Counts	2	Total Fwd/Bwd Packets
Byte Lengths	2	Total Length of Fwd/Bwd Packets
Fwd Packet Length	4	Max, Min, Mean, Std
Bwd Packet Length	4	Max, Min, Mean, Std
Throughput	2	Flow Bytes/s, Flow Packets/s
Flow IAT	4	Mean/Std/Max/Min
Fwd IAT	5	Total/Mean/Std/Max/Min
Bwd IAT	5	Total/Mean/Std/Max/Min
TCP Flags	4	Fwd/Bwd PSH, URG Flags
Flag Counts	8	FIN, SYN, RST, PSH, ACK, URG, CWE, ECE
Header Lengths	2	Fwd/Bwd Header Length
Packet Rates	2	Fwd/Bwd Packets/s
Packet Size	5	Min/Max, Mean/Std/Variance
Ratios	2	Down/Up Ratio, Average Packet Size
Bulk Rates	6	Fwd/Bwd Avg Bytes/Packets/Bulk, Rate
Subflows	4	Subflow Fwd/Bwd Packets/Bytes
Window	2	Init_Win_bytes_forward/backward
Active	4	Mean/Std/Max/Min
Idle	4	Mean/Std/Max/Min

3.3 Modbus Packet Logger

Field	Type	Description	Source
timestamp	string	ISO format UTC timestamp	modbus_logger
src_ip / dst_ip	string	Source/destination IP	modbus_logger
src_port / dst_port	int	Source/destination port	modbus_logger
direction	string	request or response	modbus_logger
function_code	int	Modbus function code (numeric)	modbus_logger
function_name	string	Human-readable function name	modbus_logger
start_address	int	Register start address	modbus_logger
values	list	Register values	modbus_logger
exception_code	string	Exception code if applicable	modbus_logger
packet_length	int	Raw byte length	modbus_logger

3.4 Topology Metrics

Metric	Type	Description	Source
num_buses	int	Number of nodes (PLCs)	topology.py
num_edges	int	Number of network links	topology.py
bus_types	list	3=slack, 1=PV, 2=PQ	topology.py
voltages	list	Per-bus voltage in p.u.	topology.py
bus_degrees	list	Per-bus connection count	topology.py
max_degree	int	Maximum bus degree	topology.py
gen_buses	list	Generator bus indices	topology.py
ip_map	dict	bus_id -> IP mapping	topology.py
depth_from_gens	list	BFS hop distance to generators	topology.py

4. Report Metrics

4.1 Executive Summary

Metric	Type	Description
total_steps	int	Total attack steps executed
detected_count	int	Steps that were detected
compromised_count	int	Steps that evaded detection
evasion_rate	float	compromised_count / total_steps
critical	int	Critical severity findings
high	int	High severity findings
medium	int	Medium severity findings
low	int	Low severity findings
info	int	Informational findings
overall_risk	string	CRITICAL / HIGH / MEDIUM / LOW

4.2 CVSS Scores

Severity	CVSS Score
Critical	9.5
High	7.5
Medium	5.0
Low	2.5
Informational	0.0

4.3 Severity Classification

Condition	Severity
Stage 3 + confidence < 0.5	Critical
Stage 3 + confidence < 0.8	High
Stage 2 + confidence < 0.3	High
Stage 2 + confidence < 0.6	Medium
Stage 1 + confidence < 0.4	Medium
Stage 1 (else)	Low
Stage 0 + confidence < 0.3	Low
Default	Informational

5. Training Metrics

5.1 PPO Hyperparameters

Parameter	Value	Description
learning_rate	3e-4	Learning rate
gamma	0.99	Discount factor
clip_range	0.2	PPO clip range
ent_coef	0.01	Entropy coefficient
n_steps	2048	Steps per update
batch_size	64	Mini-batch size
n_epochs	10	Epochs per update
net_arch	[256,256,128]	Network architecture

5.2 Evaluation Metrics

Metric	Type	Description
campaigns_completed	int	Episodes that completed all 4 stages
total_rewards	list	Per-episode total rewards
evasion_rates	list	Per-episode evasion rates
campaign_completion_%	float	completion / n_episodes * 100
avg_reward	float	Mean reward across episodes
avg_evasion	float	Mean evasion across episodes
plateau_status	string	Saturated or Progressing

6. DVS (Defense Vulnerability Score)

6.1 DVS Composite Score

Component	Weight	Formula	Range
evasion_score	30%	avg_evasion_rate * 100	0-100
campaign_score	30%	completion_rate * 100	0-100
stage_depth_score	25%	(avg_max_stage / 3) * 100	0-100
plateau_score	15%	100 if plateau, else 50	50 or 100
DVS Total	100%	Weighted sum, clipped [0,100]	0-100

6.2 DVS Rating Bands

Range	Rating	Color	Description
0-25	Robust	Green	Defense is effective
26-50	Moderate	Yellow	Some vulnerabilities exist
51-75	Vulnerable	Orange	Significant gaps in defense
76-100	Critical	Red	Defense is easily bypassed

6.3 PlateauTracker

Parameter	Default	Description
patience	5	Steps without improvement before plateau
min_evasion_gain	0.02	Min gain to reset stagnation
best_evasion_rate	0.0	Best evasion rate seen
stagnant_count	0	Consecutive episodes without improvement

7. XAI Metrics

7.1 SHAP Explainer

Metric	Description
prediction	LSTM detection confidence for the window
base_value	SHAP expected value
shap_values	2D array (WINDOW_SIZE x feature_dim)
feature_importance	Mean |SHAP| per feature across timesteps
top_features	Top-k features by importance

7.2 LIME Explainer

Metric	Description
anomaly_probability	min(reconstruction_error / (2*threshold), 1.0)
reconstruction_error	MSE for the explained row
threshold	95th percentile benign error
is_anomaly	reconstruction_error > threshold
explanation_list	LIME feature contribution weights

7.3 RL Decision Explainer

Metric	Description
chosen_probability	PPO action probability for chosen technique
action_probabilities	Full distribution over 17 techniques
value_estimate	Critic network value estimate
evasion_rate	From observation vector
last_confidence	From observation vector
completed_stages	From observation vector
reward	Step reward
detected	Step detection result
detection_confidence	Step confidence
campaign_complete	All 4 stages completed

Appendix: MITRE ATT&CK for ICS Techniques

ID	Name	Stage	Mock Prob
T0886	Remote Services	0	0.50
T0846	Remote System Discovery	0	0.70
T0888	Remote System Info Discovery	0	0.45
T0842	Network Sniffing	0	0.25
T0830	Adversary-in-the-Middle	1	0.55
T0801	Monitor Process State	1	0.35
T0802	Automated Collection	1	0.60
T0869	Standard App Layer Protocol	1	0.40
T0814	Denial of Service	2	0.85
T0878	Alarm Suppression	2	0.60
T0816	Device Restart/Shutdown	2	0.75
T0881	Service Stop	2	0.70
T0836	Modify Parameter (FDI)	2	0.65
T0835	Manipulate I/O Image	2	0.65
T0806	Brute Force I/O	2	0.80
T0832	Manipulation of View (Replay)	3	0.40
T0815	Denial of View	3	0.70